IrregFusion: A Generalized Framework for Hyperspectral Image Fusion Across Diverse Spectral Data

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***Abstract*—Fusing a low-resolution (LR) hyperspectral image (HSI) with a high-resolution (HR) multispectral image (MSI) has emerged as a promising strategy for reconstructing high-quality HSIs that combine rich spectral and fine spatial information. However, most existing HSI fusion methods operate under the restrictive assumption that the LR HSI and HR MSI are spatially aligned and fully consistent on the field-of-view (FoV), which significantly limits their applicability in real-world scenarios when such alignment is unavailable. To overcome these limitations, we propose IrregFusion, a generalized HSI fusion framework capable of handling both FoV-consistent and inconsistent fusion scenarios. In particular, IrregFusion incorporates a Transformer-based reconstruction module (illustrated in Fig. 2) that captures both intramodal and intermodal correlations between the diverse spectra data and the MSI, enhancing the model's ability to perceive and reconstruct nonlocal spectral–spatial structures. To further address the challenges posed by FoV inconsistencies, we introduce a spectral propagation strategy that diffuses observed spectral information into adjacent spectral-blank regions, thereby easing the reconstruction of missing spectral content. In addition, a self-supervised adaptation mechanism is integrated into the framework, enabling robust spectral–spatial representation learning and enhancing generalization across diverse and challenging conditions. Extensive experiments conducted on benchmark datasets demonstrate that IrregFusion effectively addresses the challenges of diverse spectral data fusion and consistently outperforms state-of-the-art (SOTA) methods in both reconstruction accuracy and visual fidelity. The source code will be released in https://github.com/JiangtaoNie/IrregFusion.git**

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I. Introduction

HYPERSPECTRAL images (HSIs) capture the reflectance of scenes across hundreds of contiguous spectral bands with narrow intervals (e.g., 10 nm) [2]. Each pixel in an HSI contains a rich spectrum that characterizes the physical reflection properties of the corresponding scene. Owing to this unique spectral discriminative power [3], HSIs have been widely used in a range of applications, including space exploration [4], military reconnaissance [5], ecological protection [6], and precision agriculture [7]. However, due to the physical limitations of spectral sensors [8], HSIs typically suffer from limited spatial resolution, which significantly degrades their performance in many downstream tasks. Consequently, extensive efforts have been devoted to HSI fusion super-resolution (SR), which aims to enhance the spatial resolution of HSIs by integrating the high-resolution (HR) spatial information in the corresponding multispectral image (MSI), and thereby facilitating their broader application in subsequent analytical tasks.

The key to HSI fusion SR methods is primarily to exploit local spectral–spatial correlations between the two modalities. Previous HSI fusion SR methods [9] typically rely on traditional optimization strategies to reconstruct the latent (i.e., HR) HSI under the guidance of handcrafted priors (such as sparsity [10], low-rank structures [11], and nonlocal similarity [8]). However, these handcrafted priors often provide limited representation capacity [12], thereby yielding suboptimal performance in challenging scenarios. Recently, deep neural network (DNN)-based methods [13], [14] have attracted increasing attention due to their strong representation ability. Earlier DNN-based approaches typically employed convolutional networks as backbones [14], [15] to learn deep representations from the input data. More recently, Transformer-based modules have demonstrated their superiority for HSI SR, primarily by leveraging self-attention mechanisms [16], [17] to effectively capture both spectral and spatial dependencies, thereby yielding richer and robust feature representations for high-resolution reconstruction.

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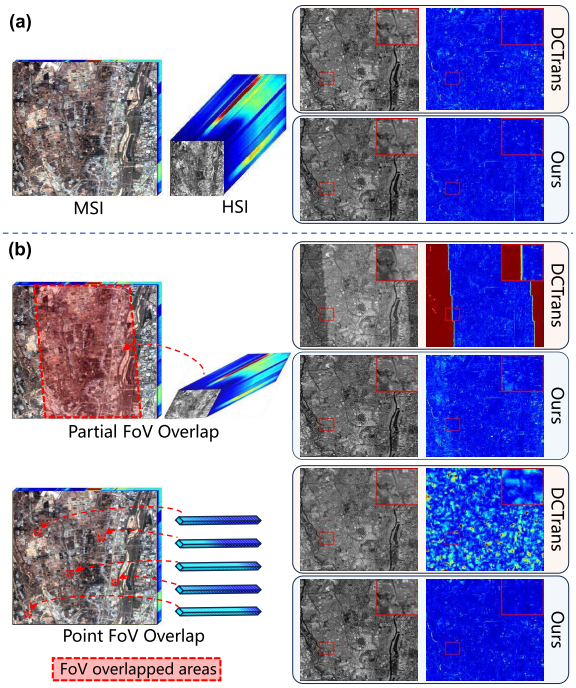


Fig. 1. Illustration of spatial information inconsistency between an HSI-MSI pair, and visual comparison of reconstruction results obtained by DCTrans [1] and the proposed IrregFusion method. (a) FoV range consistent. (b) FoV range inconsistent.

While existing HSI SR methods have demonstrated impressive fusion performance, they typically rely on a strong assumption that the low-resolution (LR) HSI and the HR MSI are spatially consistent and fully aligned [as illustrated in Fig. 1(a)]. However, this assumption is often violated in real-world remote sensing scenarios, where HSIs and MSIs are usually acquired by different imaging sensors. Due to sensor limitations and acquisition constraints, the field-of-view (FoV) of HSIs and MSIs can differ significantly, leading to spatial inconsistencies between the two modalities. In particular, since MSI sensors generally possess a wider FoV than HSI sensors, the spatial region covered by the HSI is typically a subset of the MSI and may exhibit irregular patterns, such as rectangular, parallelogram-shaped, or sparsely distributed point spectra, depending on the FoV overlap between the sensor footprints. As illustrated in Fig. 1(b), we categorize such spatial inconsistencies into two types: partial FoV overlap and point FoV overlap. Partial FoV overlap is primarily caused by misalignment between the FoV of two independent sensors. Point FoV overlap, on the other hand, refers to sparsely and irregularly distributed spectral samples [18], [19], which offer a practical and cost-effective means to obtain high-quality spectral data through reduced sampling and faster acquisition. In this study, we collectively refer to these two scenarios as irregular spectral data fusion. As further demonstrated in Fig. 1, such irregularity poses significant challenges to conventional fusion methods, which typically assume full spatial overlap and dense spectral coverage.

To address the challenges arising from spatial FoV inconsistency, we propose IrregFusion, a generalized HSI fusion framework specifically designed to accommodate various forms of spectral observations. At the core of IrregFusion is a Transformer-based reconstruction module (illustrated in Fig. 2) composed of two parallel branches that operate in a dual fashion. The first branch reconstructs missing spatial information from the degenerated spectral measurements, guided by the spatial structure provided by the HR MSI. Conversely, the second branch recovers spectral information from the HR MSI under the guidance of the available spectral information. Both branches incorporate self-attention and cross-modality attention mechanisms to effectively model intramodal and intermodal correlations between the MSI and the spectral data, thereby enhancing the model's ability to capture and reconstruct nonuniform spectral–spatial structures. To further mitigate the performance degradation caused by FoV inconsistency, we introduce a spectral propagation strategy, which diffuses the observed spectral information into nearby regions lacking spectral coverage. This not only enhances spectral continuity but also reduces the reconstruction difficulty in unobserved areas. Finally, we integrate a self-supervised adaptation mechanism into the framework, enabling effective spectral–spatial representation learning and enhancing the model's robustness and generalization across diverse and challenging conditions.

In summary, this study mainly contributes to the following three aspects.

1) We propose IrregFusion, a generalized HSI fusion framework that leverages a dual-branch Transformer-based reconstruction module to address the under-explored problem of fusing various forms of spectral observations.
2) We design a spectral propagation strategy that diffuses sparse spectral measurements into neighboring spectral-blank regions, thereby alleviating FoV inconsistency and improving reconstruction performance in areas with limited spectral coverage.
3) We incorporate a self-supervised adaptation mechanism to facilitate robust spectral–spatial representation learning, enabling the framework to generalize effectively across diverse and challenging conditions.

II. Related Works

Existing fusion-based HSI SR methods can be broadly categorized into traditional model-based approaches and deep learning-based approaches. Traditional methods typically rely on handcrafted shallow priors (e.g., sparsity [20] and low-rank [11]) combined with predefined degradation models, and formulate the fusion process as an energy minimization problem to estimate the latent HR HSI. While such priors offer good interpretability, their limited representation capacity hampers their ability to capture complex spatial–spectral patterns. As a result, these methods often fail to generalize effectively under challenging or highly variable conditions.

Recently, DNN-based HSI fusion SR methods [13], [21] have garnered significant attention due to their powerful representation capabilities. These methods can be broadly categorized into three groups: deep convolutional neural network (DCNN)-based methods [15], [22], generative model-based

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methods [23], and Transformer-based methods [1], [24], [25], [26]. DCNNs excel at extracting local structures and texture features, making them particularly effective for modeling spatially consistent patterns. For instance, Dian et al. [27] proposed a residual learning-based DCNN to learn image priors and guide the reconstruction of HR HSIs. Zhang et al. [12] introduced the UTAL framework, which leverages DCNNs to extract both image content-agnostic and image-specific priors, demonstrating strong generalization across diverse and challenging scenarios. In addition, Xie et al. [28] proposed MHF-Net, which incorporates a deep unfolding strategy into a DCNN backbone to improve model interpretability while maintaining strong fusion performance.

In contrast to DCNN-based methods that primarily focus on capturing local textures, generative model-based approaches excel at learning global semantic features and abstract representations. Shi et al. [29] proposed a latent encoder-coupled generative adversarial network (LE-GAN), which maps the generated spectral–spatial features from the image space to a latent space and introduces a coupling component to regularize the reconstructed HSI. Zhu et al. [30] integrated generative adversarial networks (GANs) with implicit neural representations to enhance the fidelity of fused images. Leveraging the strong generative capability of diffusion models, Li et al. [31] developed a dual conditional diffusion framework that generates high-quality and realistic details through iterative denoising, thereby improving reconstruction performance. More recently, Zhu et al. [23] proposed a self-learning adaptive residual-guided subspace diffusion framework that relies solely on observed images without requiring additional training data, making it particularly effective in scenarios with limited HSI samples.

Due to the Transformer model's strong capability in capturing contextual and global semantic features, it has been increasingly adopted in the field of HSI fusion SR. Ma et al. [1] proposed a dual cross-Transformer fusion framework that leverages directional pairwise multihead cross-attention to effectively model interactions between different modalities. He et al. [24] introduced LGCT, a local–global collaborative network based on Transformers, specifically designed for high-fidelity HSI reconstruction. In constructing the fusion module, Sun et al. [32] utilized the Transformer's ability to capture nonlocal information and proposed a multiscale deep cross-fusion Transformer to enhance fusion performance. Cao et al. [33] developed a cross-domain-aware Transformer (CaFormer) that integrates cross-domain attention and Fourier-domain perception to jointly model spatial–spectral dependencies and enhance multistage feature aggregation. In addition, Wang et al. [26] developed a spatial resolution-independent unfolding algorithm based on the Transformer architecture, demonstrating strong generalization across varying image resolutions and improved interpretability. Considering the influence of spatial degeneration, Cao et al. [34] proposed an unsupervised hybrid Transformer–CNN network (uHNTC) for blind HSI fusion without requiring prior degradation information.

Despite the remarkable progress achieved by the aforementioned methods, they all share a fundamental limitation: the assumption that spectral and spatial information are consistently and densely distributed across modalities. This assumption often fails in real-world scenarios, where HSI and MSI data frequently exhibit spatial inconsistency or the HSI data are sparsely and irregularly sampled (e.g., partially observed regions or point-wise spectra). As a result, their performance deteriorates significantly under such challenging conditions. These limitations underscore the necessity for a more general and robust fusion framework capable of effectively handling irregular spectral data. In contrast, the proposed IrregFusion framework is specifically designed to address these issues. By explicitly modeling spatial–spectral inconsistencies, it consistently delivers superior performance in both reconstruction accuracy and visual fidelity compared to existing state-of-the-art (SOTA) methods.

III. Methodology

A. Revisiting the HSI Fusion Method

The aim of HSI fusion method is to reconstruct the latent HR HSI, denoted by $\mathbf{Z} \in \mathbb{R}^{B\times H\times W}$, from the observed LR HSI, $\mathbf{X} \in \mathbb{R}^{B\times h\times w}$, and the HR MSI, $\mathbf{Y} \in \mathbb{R}^{b\times H\times W}$, where B denotes the number of spectral bands, and H and W represent the height and width, respectively, with $b \ll B$, $h \ll H$, and $w \ll W$. In most previous methods [10], [11], the relationships among $\mathbf{Z}$, $\mathbf{X}$, and $\mathbf{Y}$ are defined as follows:

$$\mathbf{X} = (\mathbf{k} * \mathbf{Z})\downarrow_s + \mathbf{N}_x; \quad \mathbf{Y} = \mathbf{P}\mathbf{Z} + \mathbf{N}_y \tag{1}$$

where $\mathbf{k}$ denotes the blur kernel and $\downarrow_s$ indicates the downsampling operation with a scale factor s. $\mathbf{P}$ is the spectral response function of a specific imaging sensor and is often given beforehand. $\mathbf{N}_x$ and $\mathbf{N}_y$ separately indicates the noise in observed LR HSI $\mathbf{X}$ and HR MSI $\mathbf{Y}$.

B. General HSI Fusion Framework

In (1), the LR HSI and HR MSI are assumed to share consistent spatial information, which may not hold in many real-world scenarios encountered in the HSI fusion problem. In this study, we address a more general setting, i.e., the FoV-inconsistent HSI fusion problem, in which the spectral data are irregularly distributed in space. To account for this, we propose a more flexible formulation, as defined below

$$\mathbf{X} = (\mathbf{k} * (\mathbf{M} \odot \mathbf{Z}))\downarrow_s + \mathbf{N}_x; \quad \mathbf{Y} = \mathbf{P}\mathbf{Z} + \mathbf{N}_y \tag{2}$$

where $\mathbf{M}$ is a binary mask that indicates the overlapped and nonoverlapped FoV regions and $\odot$ is the element-wise multiplication. When all elements of $\mathbf{M}$ are set to 1, the formulation corresponds to the spatially FoV-consistent fusion setting; otherwise, it represents a FoV-inconsistent fusion scenario. Since the MSI sensor typically covers a much wider FoV, in (2) we presume that the MSI shares consistent spatial information with the latent HR HSI, while the spectral information in the LR HSI is irregularly distributed. In this study, we assume that HSIs and MSIs are spatially registered within their overlapped FoV.

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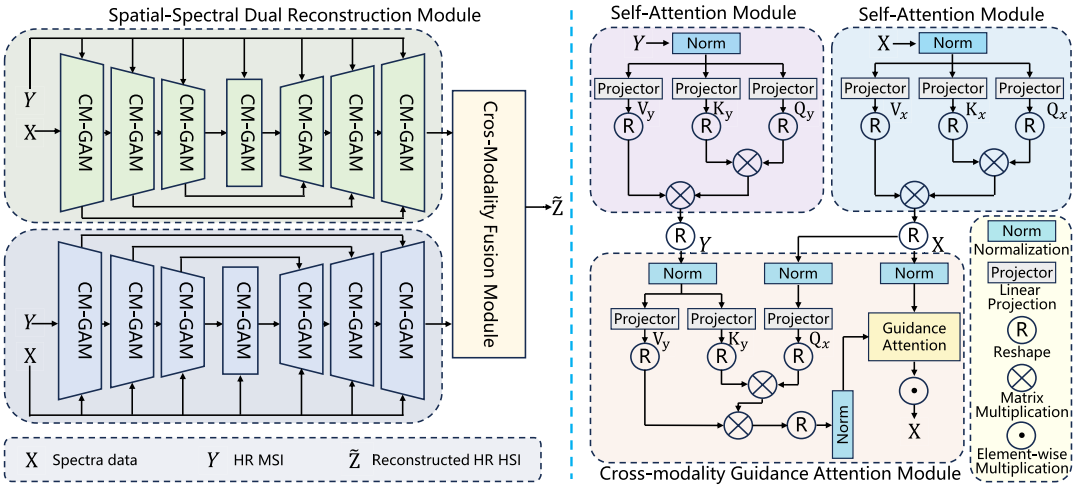


Fig. 2. Flowchart of the proposed spatial–spectral dual reconstruction module.

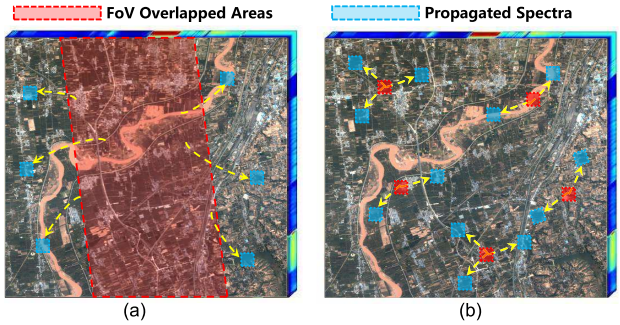


Fig. 3. Illustration of the proposed spectral propagation strategy. (a) Partial FoV overlap. (b) Point FoV overlap.

C. Spectral Propagation Strategy

As previously discussed, due to the spatial FoV inconsistency between the HR MSI and the LR HSI, some regions lack spectral information, which degrades the fusion performance. To address this limitation, we propose a spectral propagation strategy that utilizes adjacent spectral information to fill in missing spectral regions, thereby reducing the difficulty of fusion. An intuitive illustration of this strategy is shown in Fig. 3. In particular, we first compute the spatial coordinates of each pixel in both the LR HSI and the HR MSI. Subsequently, we calculate a propagation weight for each pixel based on its coordinates and apply these weights to integrate spectral information from neighboring pixels, effectively compensating for spectral deficiency and improving the robustness of the fusion process. This process can be expressed as

$$\bar{\mathbf{X}}_{i,j} = \sum_{u=-r}^{r}\sum_{v=-r}^{r} \omega_{u,v}^{i,j} \cdot \mathbf{X}_{i+u,j+v}$$
$$\text{s.t.,}\ \ \omega = \mathcal{S}\left(r - \mathcal{T}_{[0,r]}\left(\mathbf{D}\right)\right), \quad \mathbf{D} = \left\|\mathbf{C}_x - \mathbf{C}_y\right\|_2 \tag{3}$$

where $\bar{\mathbf{X}}_{i,j}$ denotes the propagated spectral vector at spatial location $\{i, j\}$, ω denotes the propagation weight computed based on the coordinate distance matrix $\mathbf{D}$, and $\mathbf{C}_x \in \mathbb{R}^{n\times 2}$ and $\mathbf{C}_y \in \mathbb{R}^{N\times 2}$ denote two coordinate matrices that record the spatial location information i, j of each pixel in $\mathbf{X}$ and $\mathbf{Y}$, respectively, where n and N denote the total number of pixels in $\mathbf{X}$ and $\mathbf{Y}$. In particular, the values in the coordinate distance matrix $\mathbf{D}$ are first truncated to lie within $[0, r]$ using a truncation function $\mathcal{T}_{[0,r]}(\cdot)$, after which a SoftMax operation $\mathcal{S}(\cdot)$ is applied to normalize the weights to range within $[0, 1]$ and sum to unity. This process enables each spectral vector $\mathbf{X}_{i,j}$ to effectively aggregate information from its neighboring spectral samples, thereby compensating for missing spectral information and alleviating the complexity of the reconstruction task.

D. Spatial–Spectral Dual Reconstruction Module

The key to HSI fusion lies in effectively exploiting the spatial–spectral relationships between the spectral data and the HR MSI. However, effectively capturing these relationships is particularly challenging due to the FoV inconsistency between spectral and spatial information in the spatial domain. A promising approach to address this challenge is to leverage cross-modality nonlocal information, thereby utilizing globally similar spectral patterns to compensate for missing spectral content. To this end, we propose a spatial-spectra dual reconstruction module (illustrated in Fig. 2) that incorporates a cross-modality guidance attention module (CM-GAM) to facilitate the reconstruction and complementation of spectral–spatial information by effectively exploiting cross-modality nonlocal dependencies.

1) Cross-Modality Guidance Attention Module: The CM-GAM comprises two key components: an intramodality self-attention module, which captures internal similarities within each modality to enhance modality-specific representation, and a CM-GAM, which exploits the intermodal complementarity to compensate for missing information and facilitate more accurate reconstruction. Initially, the features from both modalities are embedded into image tokens. For notational simplicity, we continue to denote the tokenized spectral and MSI features as $\mathbf{X}$ and $\mathbf{Y}$, respectively. Then, these tokenized features are subsequently projected into Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value ($\mathbf{V}$) representations through a series of linear projection functions denoted by $\mathcal{P}(\cdot)$. Taking spectra

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data $\mathbf{X}$ as an example, this projection process can be formulated as

$$\mathbf{Q}_x = \mathcal{P}_q^x(\mathbf{X}), \quad \mathbf{K}_x = \mathcal{P}_k^x(\mathbf{X}), \quad \mathbf{V}_x = \mathcal{P}_v^x(\mathbf{X}). \tag{4}$$

Then, we base on the project features to calculate the intrasimilarity and reconstruct the feature of each modality in a self-attention manner. Taking spectra data $\mathbf{X}$ as an example, this projection process can be formulated as

$$\mathbf{X} = \mathcal{S}\left(\frac{\mathbf{Q}_x\mathbf{K}_x^T}{\sqrt{d}} + \mathbf{P}\right)\mathbf{V}_x \tag{5}$$

where $\sqrt{d}$ is a scalar factor, $\mathbf{P}$ indicates the learnable positional encoding. After enhancing the feature representations of both the spectra data and MSI independently through the intramodality self-attention module by capturing internal similarities, we further employ the cross-modality guidance attention to exploit intermodal complementarity. To provide an intuitive understanding, we illustrate the cross-modality guidance attention using an example where the spectral features $\mathbf{X}$ are reconstructed under the guidance of the MSI features $\mathbf{Y}$. This process can be formulated as

$$\mathbf{X} = \frac{\mathbf{X}\cdot\mathbf{Y}}{\|\mathbf{X}\|\,\|\mathbf{Y}\|}\mathbf{X}, \quad \text{s.t.,}\ \mathbf{Y} = \mathcal{S}\left(\frac{\mathbf{Q}_y\mathbf{K}_x^T}{\sqrt{d}} + \mathbf{P}\right)\mathbf{V}_y. \tag{6}$$

In the CM-GAM, we first apply cross-attention to capture the complementary information between the spectra data and the MSI. The extracted cross-modal spectral representation from $\mathbf{Y}$ is then utilized as modality-specific guidance to facilitate the reconstruction of MSI features. Notably, this module is general and can also be employed in the reverse direction, that is, to reconstruct MSI features under the guidance of spectra data.

2) Dual Reconstruction Mechanism: To effectively exploit the spatial–spectral complementarity between the LR HSI and HR MSI for improved fusion performance under irregular spectral distributions, we propose to construct the IrregFusion upon a dual reconstruction mechanism. In particular, two U-shaped reconstruction modules are designed based on the proposed CM-GAM. These modules are coupled in a dual fashion, enabling each modality to extract features under the guidance of the other. Finally, the output features from both branches are fused and projected to generate the reconstructed HR HSI. This process can be formulated as

$$\tilde{\mathbf{Z}} = \mathcal{F}\left(\mathcal{U}_x(\mathbf{X},\mathbf{Y}), \mathcal{U}_y(\mathbf{Y},\mathbf{X})\right) \tag{7}$$

where $\mathcal{U}_x(\cdot)$ and $\mathcal{U}_y(\cdot)$ are the U-shaped reconstruction module for spectra data and HR MSI, respectively. $\mathcal{F}(\cdot)$ indicates the fusion module that maps the fused features into the reconstructed HR HSI. In this dual fashion, one branch recovers the missing spatial information from the irregular spectra data under the guidance of the HR spatial information provided by the HR MSI. Conversely, the other branch reconstructs spectral details from the HR MSI with the aid of the irregular spectra data. Through this bidirectional guidance, cross-modal complementarity is fully leveraged to compensate for missing information, thereby enabling more accurate and robust reconstruction.

E. Self-Supervised Adaptation Mechanism

Compared with conventional HSIs, irregular spectra data often exhibit diverse and variable spectral distributions, which can hinder the adaptability of methods with limited generalization capacity, resulting in significant performance degradation. To address this issue, we introduce a self-supervised adaptation mechanism aimed at enhancing the generalization ability of the proposed IrregFusion framework. In particular, we propose to optimize a lightweight adaptation network that refines the reconstruction quality under a self-supervised constraint derived from the observed irregular spectra data. A promising approach to achieve this self-supervised learning scheme is to construct a maximum a posteriori (MAP) estimation similar to [12], which can be expressed by optimizing the following objective:

$$\begin{aligned} &\min_{\theta_a}\ \alpha\left\|\mathbf{Y} - \mathbf{P}\hat{\mathbf{Z}}\right\|_2 + \beta\left\|\mathbf{X} - \left(\mathbf{k} * \hat{\mathbf{Z}}\right)\downarrow_s\right\|_2 \\ &\text{s.t.,}\ \hat{\mathbf{Z}} = \mathcal{A}\left(\tilde{\mathbf{Z}};\theta_a\right), \quad \tilde{\mathbf{Z}} = \mathcal{F}\left(\mathbf{X},\mathbf{Y};\theta_f\right) \end{aligned} \tag{8}$$

where $\mathcal{F}(\cdot;\theta_f)$ is the reconstruction network with θ_f as its parameter, $\mathcal{A}(\cdot;\theta_a)$ indicates the self-supervised adaptation module parameterized with θ_a. However, due to the irregular nature of the spectra data in the LR HSI and its spatial inconsistency with the latent HR HSI, the observed LR HSI may fail to provide accurate and complete spectral information across the entire scene. To address this challenge, we propose a more general self-supervised adaptation mechanism. We introduce an additional constraint to strengthen supervision in the FoV nonoverlapped regions, effectively correcting the spectral information in these areas and thereby achieving significant improvements in fusion quality. This mechanism treats the observed pure spectra data as strong supervision and leverages the predicted spectral information as a weak supervisory signal. By integrating both sources, it establishes a robust self-supervised spectral constraint that effectively covers the entire spatial domain. This mechanism can be formulated as optimizing the following problem:

$$\begin{aligned} &\min_{\theta_a}\ \alpha\left\|\mathbf{Y} - \mathbf{P}\hat{\mathbf{Z}}\right\|_2 + \beta\left\|\mathbf{X}_m - \hat{\mathbf{X}}_m\right\|_2 + \gamma\left\|\tilde{\mathbf{X}}_{\bar{m}} - \hat{\mathbf{X}}_{\bar{m}}\right\|_2 \\ &\text{s.t.,}\ \hat{\mathbf{Z}} = \mathcal{A}\left(\tilde{\mathbf{Z}};\theta_a\right), \quad \tilde{\mathbf{Z}} = \mathcal{F}\left(\mathbf{X},\mathbf{Y};\theta_f\right) \end{aligned} \tag{9}$$

where $\mathbf{X}_m \in \mathbb{R}^{n\times B}$, $\hat{\mathbf{X}}_m \in \mathbb{R}^{n\times B}$, and $\tilde{\mathbf{X}}_m \in \mathbb{R}^{n\times B}$ denote the observed spectra data, and the spatial degenerated spectra outputs from the adaptation module and the reconstruction network, respectively. All three data representations are masked by the spectral distribution mask $\mathbf{M}$ to eliminate the blank regions. Meanwhile, $\bar{\mathbf{M}} = 1 - \mathbf{M}$ indicates the areas with missing spectral information. By incorporating both strong supervision from the observed spectra data and weak supervision from the predicted outputs, the proposed framework effectively guides the model to adapt to varying spectral distributions, thereby improving its generalization capability. As illustrated in Fig. 4, the self-supervised adaptation module consists of several convolutional layers combined with a self-attention mechanism, enabling it to fully exploit the intrinsic similarity within each test sample.

Moreover, to further clarify the methodology, we provide two algorithms that summarize the entire IrregFusion

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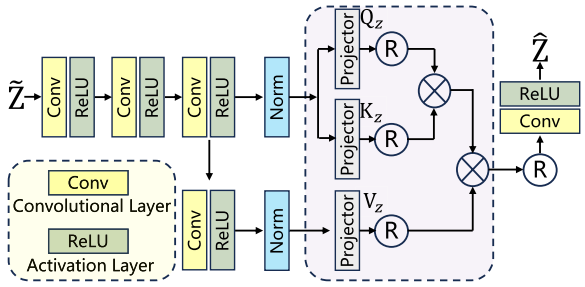


Fig. 4. Illustration of the proposed self-supervised adaptation module.

Algorithm 1 Training Procedure of the Spatial–Spectral Dual Reconstruction Module

Input: LR HSI $\mathbf{X}$, HR MSI $\mathbf{Y}$, mask $\mathbf{M}$;
Initialization: Initialize backbone network $\mathcal{F}(\cdot;\theta_f)$ with random parameters;
Output: Fusion result $\tilde{\mathbf{Z}}$;
for each epoch = 1 to 150 **do**
    Construct coordinate $\mathbf{C}_x$ and $\mathbf{C}_y$ for $\mathbf{X}$ and $\mathbf{Y}$;
    Compute propagated spectral data $\bar{\mathbf{X}}$ using the spectral propagation strategy (Eq. (3));
    Feed $\bar{\mathbf{X}}$ and $\mathbf{Y}$ into $\mathcal{F}(\cdot;\theta_f)$ to obtain the output $\tilde{\mathbf{Z}}$;
    Calculate the training loss and update parameters θ_f via backpropagation;
end for

framework. In particular, Algorithm 1 presents the training procedure of the proposed spatial–spectral dual reconstruction module, while Algorithm 2 describes the inference procedure applied to test samples. During the inference phase, the parameters θ_f of the reconstruction network $\mathcal{F}(\cdot;\theta_f)$ are kept frozen, and only the self-supervised adaptation module is optimized. For each test sample, this module is randomly initialized and trained under the supervision of (9), producing a sample-specific adaptation output.

IV. Experiments

A. Experimental Settings

1) Datasets: In this study, we evaluate the proposed method on four HSI fusion SR datasets, comprising three RS datasets and one natural scene dataset: GF5 [35], Pavia [36], CAVE [37], and HypSen [38].

a) GF5 Dataset: The GF5 dataset includes four HSIs of size 2083 × 2008 with 150 spectral bands, spanning the wavelength range of 400–1000 nm at 5 nm intervals. For practical use, each HSI is cropped into nonoverlapping 512 × 512 subimages, retaining only those with clear and distinct structural content. Subimages from the first three HSIs are used as the training set (12 samples), while those from the fourth HSI serve as the test set (nine samples).

b) Pavia Dataset: The Pavia dataset consists of two HSIs captured by the ROSIS over the city of Pavia, covering the wavelength range from 430 to 860 nm. The images represent two scenes: Pavia Center (1096 × 715) and Pavia University (610 × 340). After discarding noisy and water absorption bands, 102 spectral bands are retained. The Pavia Center HSI is divided into 256×256 patches for training, while the Pavia University HSI is used exclusively for testing.

Algorithm 2 Inference Procedure of the IrregFusion Framework

Input: LR HSI $\mathbf{X}$, HR MSI $\mathbf{Y}$, mask $\mathbf{M}$;
Initialization: Pretrained spatial–spectral dual reconstruction module $\mathcal{F}(\cdot;\theta_f)$; initialize self-supervised adaptation module $\mathcal{A}(\cdot;\theta_a)$ with random parameters;
Output: Final fusion result $\hat{\mathbf{Z}}$;
Preprocessing:
    (i) Construct coordinate $\mathbf{C}_x$ and $\mathbf{C}_y$ for $\mathbf{X}$ and $\mathbf{Y}$;
    (ii) Compute propagated spectral data $\bar{\mathbf{X}}$ using the spectral propagation strategy (Eq. (3));
    (iii) Feed $\bar{\mathbf{X}}$ and $\mathbf{Y}$ into the pretrained $\mathcal{F}(\cdot;\theta_f)$ to obtain the preliminary fusion output $\tilde{\mathbf{Z}}$; note that θ_f remains frozen during inference;
Self-supervised Adaptation:
for epoch = 1 to 500 **do**
    Feed $\tilde{\mathbf{Z}}$ into $\mathcal{A}(\cdot;\theta_a)$ to generate the adapted result $\hat{\mathbf{Z}}$;
    Compute the self-supervised adaptation loss (Eq. (9)) and update θ_a via backpropagation;
end for
Return: Final fusion result $\hat{\mathbf{Z}}$ obtained at the 500th epoch.

c) CAVE Dataset: The CAVE dataset comprises 32 HSIs primarily depicting indoor scenes, each with a spatial resolution of 512×512 and 31 spectral bands covering 400–700 nm at 10 nm intervals. We randomly select 20 HSIs for training and use the remaining 12 as the test set.

d) HypSen Dataset: The HypSen dataset is a real-world dataset comprising an HSI acquired by the Hyperion sensor onboard the Earth Observing-1 (EO-1) satellite and an MSI captured by the Sentinel-2A satellite. Both the HSI and MSI depict the same geographical region in Lafayette, Louisiana, USA, and were collected separately in October and November 2015. The HSI has a spatial resolution of 30 m and includes 242 spectral bands spanning the wavelength range from 400 to 2500 nm. The MSI has a higher spatial resolution of 10 m and consists of six spectral bands. For evaluation purposes, we extract a 100×100 subregion from the HSI and a corresponding 300×300 subregion from the MSI as the test sample.

In the experiments, except for the HypSen dataset, each HSI in the other three datasets is treated as the ground truth of the latent HR HSI $\mathbf{Z}$. The corresponding LR HSI $\mathbf{X}$ and HR MSI $\mathbf{Y}$ are synthesized from $\mathbf{Z}$ using the degradation functions described in (2). In particular, following the settings in [10] and [11], spatial degradation is implemented via a single convolutional layer with zero bias and a stride of s. The convolutional kernel is fixed as an 8×8 Gaussian kernel $\mathbf{k}$ with zero mean and a variance of 3. The stride parameter s determines the down-sampling scale applied to the input and is set to 8 in this work. For spectral degradation, the spectral response functions specific to each dataset are applied to generate the HR MSI $\mathbf{Y}$ from the HR HSI $\mathbf{Z}$. The resulting HR MSI maintains the same spatial size as the target HR HSI.

To simulate FoV inconsistent in synthetic datasets (e.g., GF5, Pavia, and CAVE), we randomly generate masks $\mathbf{M}$.

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Considering the imaging characteristics of HSI and MSI sensors, we assume that the MSI covers a wider FoV, while the HSI captures a relatively smaller one. Based on the generated masks, we simulate FoV-inconsistent HSI data. Two types of inconsistencies are considered: partial FoV overlap and point FoV overlap. In the partial overlap case, the HSI is restricted to a continuous spatial region with a smaller FoV. For simulation, the overlapping region is randomly shaped as a rectangle, parallelogram, or ellipse, and the coverage ratio is varied (e.g., 70%, 50%, and 30%) to model different levels of FoV overlapping. In the point overlap case, spectral information is retained only in sparse, discrete locations. This is achieved by applying random binary masks with different sparsity ratios (e.g., 1%, 0.5%, and 0.1%), where, for example, a 1% mask indicates that spectral vectors are available for only 1% of the total pixels in the spatial domain of the latent HR HSI.

2) Implementation Details: The proposed IrregFusion framework primarily comprises two components: a spatial –spectral dual reconstruction module and a self-supervised-based adaptation module. During the training phase, we exclusively train the spatial–spectral dual reconstruction module to fully exploit both intramodality and intermodality spatial–spectral correlations between the irregular spectra data and the HR MSI. In the inference phase, this module is fixed (i.e., not updated), and only the adaptation module is optimized using the self-attention mechanism to further enhance fusion performance. To enable the spatial–spectral dual reconstruction module to generalize across different types of spectral inputs, we adopt a synthetic training strategy. In particular, based on the mask types described in (2), we dynamically generate various forms of spectral degradation. For instance, a mask filled with ones indicates the FoV of LR HSI is consistent with the latent HR HSI, whereas masks with values of one only in closed regions (e.g., rectangles and parallelograms) or random pixel locations simulate irregular spectral conditions. The training process is implemented using the PyTorch framework. We adopt the Adam optimizer [39] with an ℓ_1 norm-based loss function. The training process is conducted over 150 epochs on three NVIDIA GeForce 2080ti GPUs. The initial learning rate is set to 1×10^{-4}, and the batch size is set to 6. During the inference phase, we focus on optimizing the adaptation module in a self-supervised manner, where only the observed irregular spectra data and the HR MSI are utilized for network optimization. The adaptation process is implemented using the PyTorch framework, with the Adam optimizer initialized at a learning rate of 1×10^{-3}. We optimize the objective function defined in (9), employing an ℓ_1 norm-based loss function. For simplicity and empirical effectiveness, the balance weights α, β, and γ are set to 1, 1, and 0.1, respectively. The optimization process is terminated after 500 iterations.

3) Comparison Methods: In this study, we compare the proposed method with nine SOTA HSI reconstruction approaches: LGCT [24], SPECAT [25], ESSA [40], DCTrans [1], CLoRF [41], OTIAS [42], MCIFNet [43], UTAL [12], and PSTUN [26]. In this study, we assess each method under its default training configuration as described in the original publication to examine its effectiveness in handling irregular spectral fusion. To quantitatively evaluate reconstruction performance, we adopt four widely used metrics: root mean square error (RMSE), peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), and structural similarity index measure (SSIM). In general, higher PSNR and SSIM values, along with lower RMSE and SAM values, indicate better reconstruction quality.

B. Performance Evaluation

In this section, we present a comprehensive comparison between the proposed IrregFusion framework and several SOTA methods across three benchmark datasets. In particular, we simulate different kinds of spectral inconsistency to assess the robustness and effectiveness of existing methods in irregular spectral fusion scenarios, thereby emphasizing the advantages of our approach. In Section IV-B, we provide a detailed description and analysis of the experimental results for both settings.

1) Performance Evaluation on the GF5 Dataset: In this article, we conduct a series of comparative experiments on the GF5 datasets under settings with spatial information consistent and mild spectral–spatial inconsistencies (i.e., 70% partial FoV overlap and 1% point FoV overlap). In the partial FoV overlap setting, 70% represents the proportion of the LR HSI that spatially overlaps with the HR MSI in terms of FoV. For the point FoV overlap setting, where a 1% indicates that spectral vectors are retained for only 1% of the total pixels in the spatial domain of latent HR HSI. All baseline methods are implemented using their default training configurations, as described in their original publications. This set of experiments is designed to investigate two main aspects: first, the impact of spatial inconsistency on the performance of general HSI fusion methods, and second, the effectiveness of the proposed IrregFusion framework in addressing such inconsistencies.

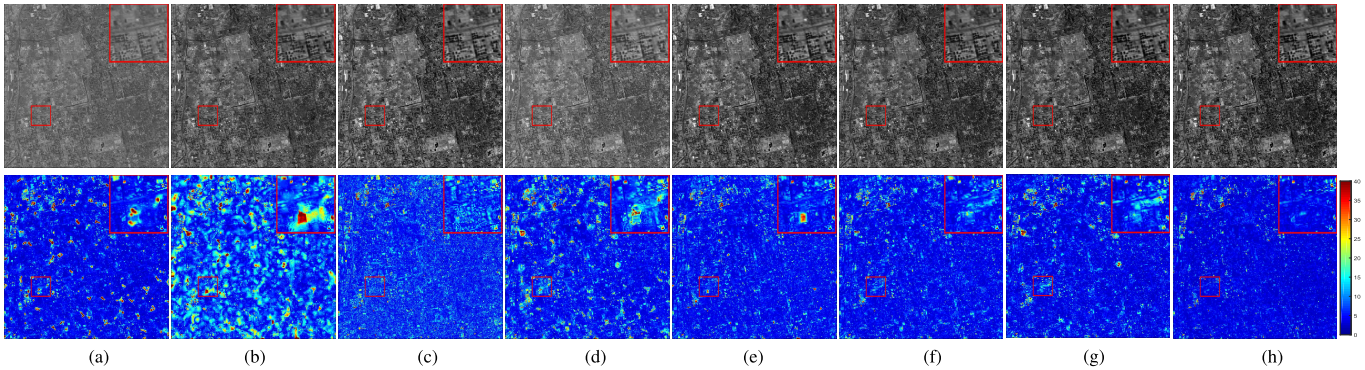
The quantitative results are summarized in Table I. As observed, all comparison methods achieve strong performance when the spatial FoV is consistent. However, their performance significantly degrades across all evaluation metrics when confronted with irregular spectra data, highlighting their limited capacity to address such challenging scenarios. In contrast, under irregular spectral settings, the proposed method consistently outperforms the second-best approach by a substantial margin of 2 to 4 dB in PSNR, demonstrating its superior effectiveness in handling irregular spectral fusion. Among these methods, it can be observed that DCTrans achieves the best performance under the FoV-consistent setting. However, its performance degrades drastically when evaluated under FoV-inconsistent conditions. A plausible explanation is that DCTrans is heavily tailored to the assumption of perfect spatial alignment between the LR HSI and HR MSI. Consequently, the model tends to overfit to the FoV-aligned setting, and once this assumption is violated, its ability to capture reliable spatial–spectral correlations is severely compromised, leading to a marked decline in performance. This phenomenon is not unique to DCTrans; rather, it reflects a common limitation of many existing HSI fusion methods, which are primarily designed under the assumption of full FoV overlap. As a result, when exposed to FoV-inconsistent

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TABLE I

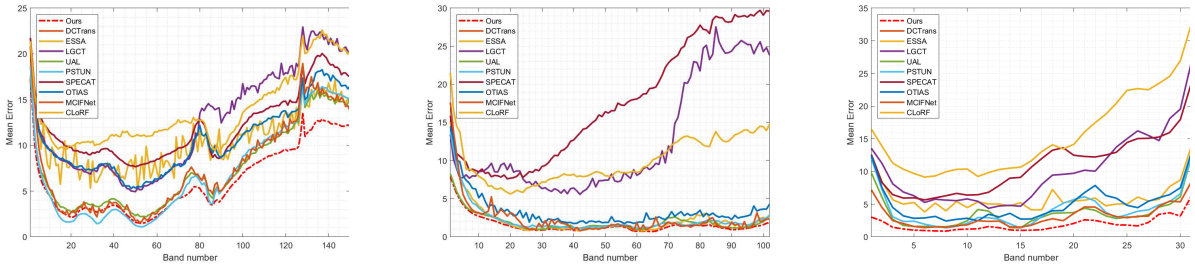
QUANTITATIVE PERFORMANCE OF EACH METHOD ON THE GF5 DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Methods	FoV Consistent				Partial FoV Overlap (70%)				Point FoV Overlap (1%)			
	RMSE↓	PSNR↑	SAM↓	SSIM↑	RMSE↓	PSNR↑	SAM↓	SSIM↑	RMSE↓	PSNR↑	SAM↓	SSIM↑
LGCT	6.84	31.53	2.62	0.9638	41.59	15.91	12.53	0.8425	14.13	25.16	4.71	0.9363
SPECAT	7.18	31.13	2.74	0.9617	30.52	18.48	6.13	0.9006	11.90	26.69	3.83	0.9387
ESSA	5.77	32.95	2.24	0.9744	20.41	22.10	7.29	0.9375	10.73	27.55	3.38	0.9467
DCTrans	5.12	33.99	2.06	0.9787	36.54	16.89	6.51	0.8828	13.90	25.31	3.92	0.9491
CLoRF	8.07	30.06	3.04	0.9468	9.07	27.01	3.43	0.9445	10.26	27.98	3.90	0.9251
OTIAS	6.74	31.64	2.64	0.9653	12.96	25.89	4.60	0.9499	10.64	27.65	3.58	0.9432
MCIFNet	6.66	31.74	2.73	0.9727	22.61	21.13	7.07	0.9405	7.90	30.24	3.07	0.9684
UTAL	5.85	32.82	2.31	0.9743	10.45	27.77	3.33	0.9631	7.31	30.89	2.64	0.9701
PSTUN	5.62	33.20	2.21	0.9756	11.02	27.30	3.74	0.9704	7.86	30.29	2.88	0.9704
Ours	5.60	33.24	2.20	0.9762	6.40	32.02	2.49	0.9741	6.29	32.25	2.38	0.9741



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 5. Visual SR results and corresponding reconstruction error maps of each method on the GF5 dataset, evaluated under the point FoV overlap setting with only 1% of spectral vectors available. (a) ESSA. (b) DCTrans. (c) CLoRF. (d) OTIAS. (e) MCIFNet. (f) UTAL. (g) PSTUN. (h) Ours.



(a) (b) (c)

Fig. 6. Mean reconstruction error curves of all methods across spectral bands on the GF5, PaviaU, and CAVE datasets. (a) GF5. (b) PaviaU. (c) CAVE.

scenarios, these methods generally fail to provide satisfactory fusion results.

Although the proposed method achieves only comparable performance under spatially consistent conditions, this is expected given that all comparison methods are trained exclusively on fully aligned data, whereas our approach is designed to balance performance across a range of spectral configurations, including both consistent and irregular distributions. In addition, visual comparisons of the reconstructed HSIs under the point FoV overlap setting (with only 1% of spectral vectors available) are presented in Fig. 5. As illustrated, most competing methods produce noticeable artifacts, particularly in regions lacking spectral information. This observation is further corroborated by the corresponding reconstruction error maps, which reveal significantly higher error concentrations in areas without spectral data. These findings suggest that the primary reason for the failure of existing methods lies in their inability to effectively address the issue of spectral modality missingness. To further substantiate these results, Fig. 6(a) presents the mean reconstruction error curves of all methods across spectral bands. It can be seen that most comparison methods suffer from high reconstruction error across spectral bands, whereas the proposed method maintains close fidelity to the ground truth.

2) Performance Evaluation on the Pavia Dataset: In this section, we conduct a series of comparative experiments on the Pavia datasets under more challenging spatial inconsistency settings, specifically with 50% partial FoV overlap and 0.5% point FoV overlap. The quantitative results are reported in Table II. Consistent with the findings on the GF5 dataset, all baseline methods perform well under the spatial

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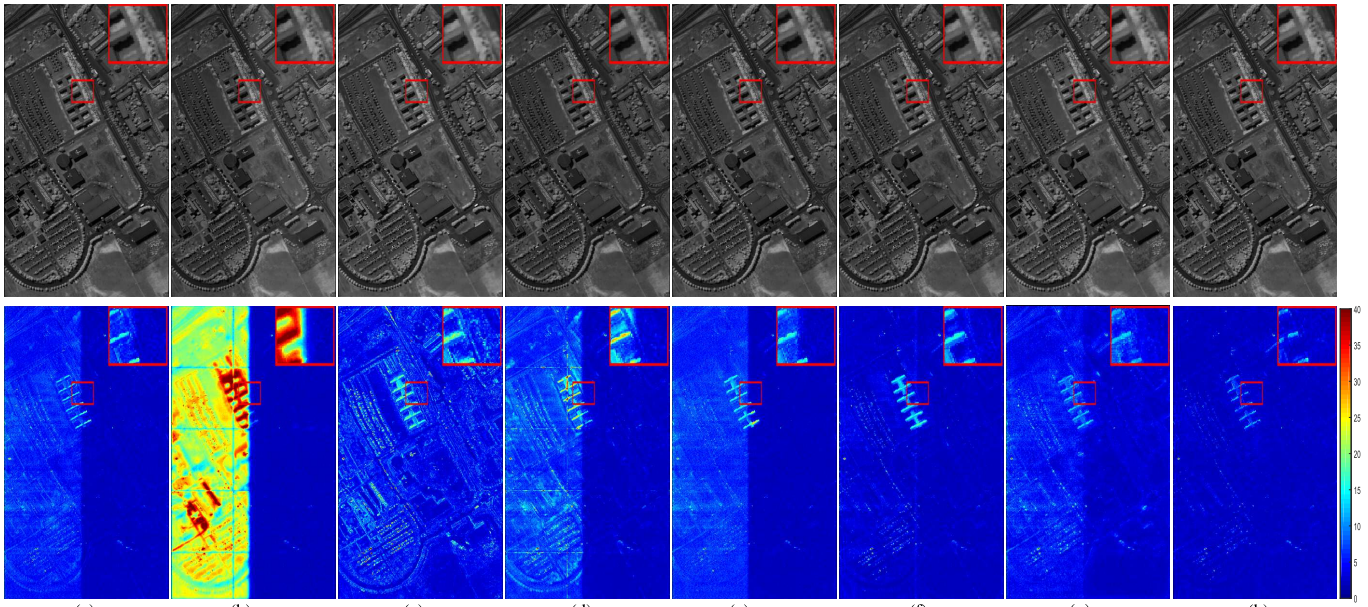


Fig. 7. Visual SR results and corresponding reconstruction error maps of each method on the Pavia University dataset, evaluated under the 50% partial FoV overlap setting. (a) ESSA. (b) DCTrans. (c) CLoRF. (d) OTIAS. (e) MCIFNet. (f) UTAL. (g) PSTUN. (h) Ours.

TABLE II

QUANTITATIVE PERFORMANCE OF EACH METHOD ON THE PAVIA UNIVERSITY DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Methods	FoV Consistent				Partial FoV Overlap (50%)				Point FoV Overlap (0.5%)			
	RMSE↓	PSNR↑	SAM↓	SSIM↑	RMSE↓	PSNR↑	SAM↓	SSIM↑	RMSE↓	PSNR↑	SAM↓	SSIM↑
LGCT	2.47	40.17	2.55	0.9771	15.58	24.29	12.79	0.8798	8.71	29.33	7.34	0.9144
SPECAT	2.35	40.60	2.40	0.9790	18.87	22.62	9.21	0.8321	7.12	31.09	4.58	0.9158
ESSA	1.86	42.52	2.07	0.9834	3.09	38.26	3.61	0.9758	5.49	33.32	4.80	0.9163
DCTrans	1.65	43.33	1.93	0.9840	10.31	27.86	6.01	0.9149	5.92	32.67	4.13	0.9548
CLoRF	3.18	38.01	3.11	0.9672	4.70	34.90	4.39	0.9300	4.23	35.57	4.00	0.9442
OTIAS	2.12	41.44	2.34	0.9804	3.81	36.46	4.08	0.9649	4.79	34.51	4.22	0.9377
MCIFNet	1.99	41.98	2.08	0.9835	3.32	37.64	3.85	0.9754	2.51	40.02	2.73	0.9790
UTAL	1.74	43.08	2.05	0.9835	2.27	40.86	2.42	0.9808	2.44	40.25	2.69	0.9793
PSTUN	1.74	43.09	2.04	0.9835	2.94	38.68	3.54	0.9750	2.89	38.85	3.48	0.9752
Ours	1.72	43.11	2.02	0.9837	1.88	42.46	2.18	0.9830	1.95	42.12	2.24	0.9826

information consistent setting. However, their performance deteriorates significantly across all evaluation metrics when confronted with spatial–spectral inconsistencies, underscoring their limited capacity to address such scenarios. In contrast, the proposed method consistently outperforms the second-best approach by a notable margin of approximately 2 dB in PSNR under irregular spectral settings, highlighting its superior effectiveness in handling challenging fusion tasks. In addition, visual comparisons of the reconstructed HSIs under the partial spectra setting (50% partial FoV overlap) are presented in Fig. 7. As shown, most competing methods produce obvious artifacts, particularly at the boundary between regions with and without spectral information. The left region lacks spectral data, while the right region contains it, resulting in distinct discontinuities. This observation is further supported by the corresponding reconstruction error maps, which show significantly higher error concentrations in areas lacking spectral data. To further validate these findings, the mean reconstruction error curves of all methods across spectral bands are plotted in Fig. 6(b). We can notice that the proposed method maintains the lowest reconstruction error across the spectral bands.

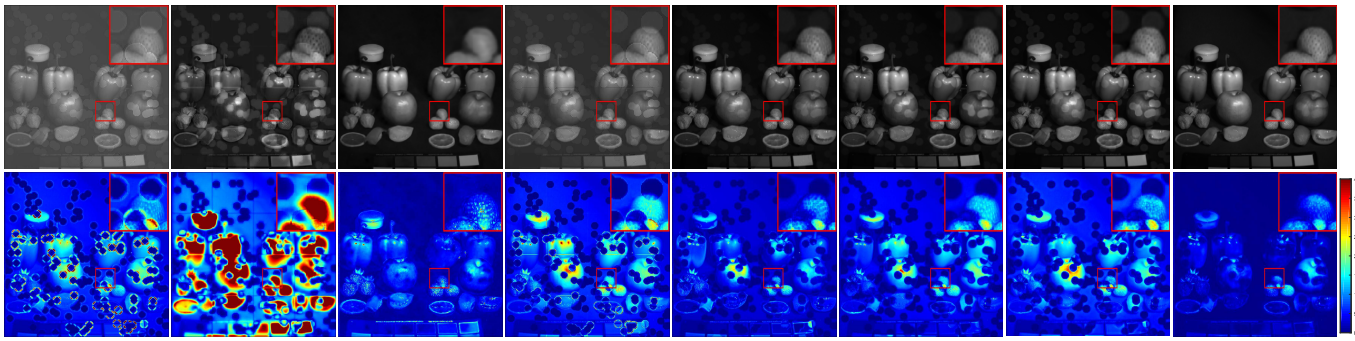
3) Performance Evaluation on the CAVE Dataset: In this section, we conduct a series of comparative experiments on the CAVE dataset to evaluate the generalization ability of the proposed method on natural scene HSIs. The experiments are carried out under highly challenging spatial inconsistency settings, specifically with 30% partial FoV overlap and 0.1% point FoV overlap. The quantitative results are presented in Table III. Consistent with the observations from the GF5 and Pavia datasets, all baseline methods perform competitively under the spatial information consistent setting. However, their performance deteriorates markedly across all evaluation metrics when faced with spatial inconsistencies, underscoring their limited capacity to address such scenarios. In contrast, the proposed method consistently outperforms the second-best approach by a substantial margin, approximately 4 dB in PSNR and 7 units in SAM, under irregular spectral con-

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TABLE III

QUANTITATIVE PERFORMANCE OF EACH METHOD ON THE CAVE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Methods	FoV Consistent				Partial FoV Overlap (30%)				Point FoV Overlap (0.1%)			
	RMSE↓	PSNR↑	SAM↓	SSIM↑	RMSE↓	PSNR↑	SAM↓	SSIM↑	RMSE↓	PSNR↑	SAM↓	SSIM↑
LGCT	1.91	43.92	3.72	0.9869	20.02	22.10	17.24	0.8175	18.13	22.96	15.83	0.7879
SPECAT	2.30	42.50	4.10	0.9829	14.65	25.56	17.44	0.9163	13.22	26.71	17.00	0.8658
ESSA	1.54	45.70	3.00	0.9919	5.67	33.94	16.40	0.9533	8.55	29.96	16.68	0.8680
DCTrans	1.23	47.57	2.24	0.9950	19.55	22.45	41.17	0.6226	18.75	22.89	34.54	0.6668
CLoRF	2.40	41.08	7.24	0.9682	6.16	33.66	14.00	0.8972	4.80	35.42	11.15	0.9325
OTIAS	1.83	44.35	3.53	0.9884	6.73	32.52	18.64	0.9409	7.55	31.31	18.72	0.9055
MCIFNet	2.26	42.82	3.66	0.9882	5.82	33.81	16.17	0.9572	5.91	33.53	15.62	0.9501
UTAL	1.51	45.61	2.95	0.9930	5.49	34.35	13.85	0.9624	5.67	33.86	14.58	0.9519
PSTUN	1.54	46.38	2.49	0.9940	6.32	33.01	19.65	0.9430	6.46	32.70	19.16	0.9351
Ours	1.40	46.24	2.67	0.9940	3.45	38.79	5.80	0.9864	3.54	38.55	5.80	0.9846



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 8. Visual SR results and corresponding reconstruction error maps of each method on the CAVE dataset, evaluated under the point FoV overlap setting with only 0.1% of spectral vectors available. (a) ESSA. (b) DCTrans. (c) CLoRF. (d) OTIAS. (e) MCIFNet. (f) UTAL. (g) PSTUN. (h) Ours.

TABLE IV

QUANTITATIVE PERFORMANCE OF DIFFERENT STRATEGIES FOR COMPENSATING THE MISSING SPECTRAL INFORMATION IN HSI, EVALUATED ON THE GF5 DATASET WITH 70% PARTIAL FOV OVERLAP BETWEEN HSI AND MSI

Methods	W/O Compensating Strategy			Implicit Neural Representation			Deep Spatial Interpolation			Inpainting			Spectral Propagation Strategy		
	PSNR↑	SAM↓	SSIM↑	PSNR↑	SAM↓	SSIM↑	PSNR↑	SAM↓	SSIM↑	PSNR↑	SAM↓	SSIM↑	PSNR↑	SAM↓	SSIM↑
ESSA	22.10	7.29	0.9375	25.51	5.96	0.9474	27.71	4.41	0.9394	29.84	3.30	0.9615	30.54	2.88	0.9674
MCIFNet	21.13	7.07	0.9405	26.02	5.44	0.9634	29.32	3.53	0.9703	30.29	3.14	0.9706	31.00	2.85	0.9724
OTIAS	25.89	4.60	0.9499	25.12	6.13	0.9449	27.96	4.14	0.9499	28.51	3.70	0.9487	29.53	3.33	0.9589
DCTrans	16.89	6.51	0.8826	22.82	5.99	0.9491	26.39	4.00	0.9628	27.21	3.46	0.9645	30.53	2.93	0.9742
PSTUN	27.30	3.74	0.9704	27.02	4.91	0.9673	30.22	3.22	0.9732	31.07	2.85	0.9739	31.45	2.78	0.9743
UAL	27.77	3.33	0.9631	29.81	3.23	0.9689	29.91	3.17	0.9715	31.00	2.74	0.9725	30.14	3.00	0.9695
Ours	30.71	3.05	0.9731	30.71	2.98	0.9714	30.68	2.93	0.9729	31.53	2.61	0.9735	32.08	2.48	0.9742

ditions, demonstrating its superior effectiveness in handling complex fusion tasks. Furthermore, visual comparisons of the reconstructed HSIs under the point FoV overlap setting (with only 0.1% of spectral vectors available) are shown in Fig. 8. As illustrated, most competing methods produce noticeable artifacts. This phenomenon is further confirmed by the corresponding reconstruction error maps, which reveal minimal error near the point spectra and significantly elevated errors in regions far from any available spectral data. To further support these findings, the mean reconstruction error curves of all methods across spectral bands are plotted in Fig. 6. Most baseline methods suffer from pronounced reconstruction errors. In contrast, the proposed method achieves spectral reconstructions that closely align with the ground truth, highlighting its robustness and precision under extreme spectral sparsity.

4) Performance Evaluation With Extra Spectral Compensating Strategies: In this section, we conducted additional experiments by introducing three alternative approaches for compensating missing spectral information in HSI: an implicit neural representation method (SQformer [44]), a deep spatial interpolation method (HSRMamba [45]), and an HSI inpainting method (MPHSIR [46]). These three methods were trained independently under identical settings to learn how to address spectral information loss caused by FoV inconsistency between HSI and MSI. The compensated LR HSI was then used as input to all fusion methods to generate the final results. In addition, we applied our proposed spectral propagation strategy to these comparison methods for a more comprehensive evaluation.

The quantitative results are summarized in Table IV. Compared with directly fusing FoV-inconsistent HSI and MSI, all

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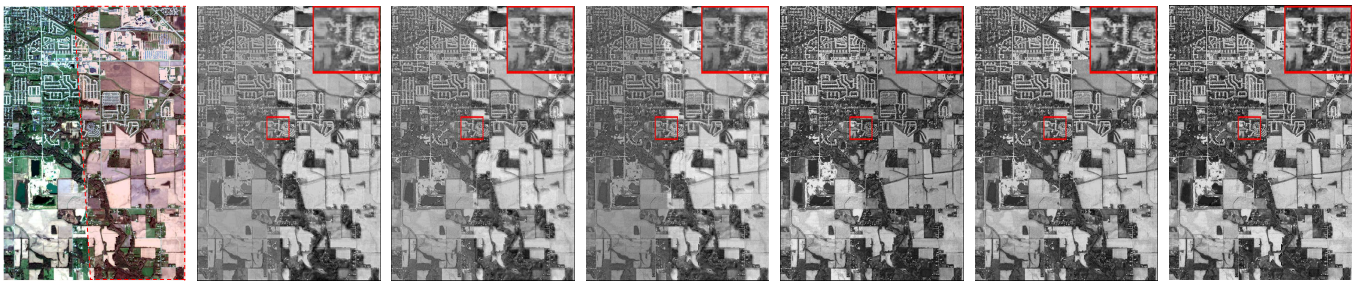


Fig. 9. Visual SR results of each method on the real-world dataset, evaluated under the partial spectra setting with a 50% spatial–spectral overlap. The region in the red box indicates the spatial–spectral overlap area. (a) HR MSI. (b) SPECAT. (c) ESSA. (d) DCTrans. (e) UTAL. (f) PSTUN. (g) Ours.

compensation strategies improve fusion performance. Among them, the implicit neural representation-based method provides the least improvement, and only for a subset of methods. We attribute this to its limited receptive field (e.g., using only adjacent 3×3 pixels for upsampling), which constrains its ability to compensate for missing spectral information when the FoV inconsistency is large. By contrast, the deep spatial interpolation method yields clear performance gains across all methods, likely because it expands the receptive field through cascaded module layers, thereby enabling more effective spectral compensation. The inpainting-based method further enhances fusion performance, outperforming the other two alternatives. However, our proposed spectral propagation strategy achieves the most significant improvement. This is because it leverages the coordinate relationships of pixels in the LR HSI to aggregate spectral information over a much broader spatial range, resulting in more effective compensation of missing spectral information.

In summary, the comparison methods are primarily designed for the spatial information consistent setting and thus perform well under such conditions. However, when confronted with spatial inconsistencies, including partial FoV overlap or irregular point spectra, these methods exhibit significant performance degradation. In contrast, the proposed IrregFusion framework is a generalized fusion approach that seamlessly integrates various spectral data configurations, ranging from fully consistent data to FoV-inconsistent data, within a unified architecture. It consistently achieves competitive performance across all scenarios, demonstrating its strong adaptability and robustness to irregular spectral fusion challenges.

C. Performance Evaluation on the Real Dataset

Furthermore, we evaluate all methods on a real-world dataset characterized by 50% partial FoV overlap. The visual comparison results are presented in Fig. 9, where regions with available spectral data are highlighted by red boxes on the HR MSI image. While the comparison methods exhibit promising performance on synthetic datasets with spatially consistent data, their reconstructions on spatially inconsistent real-world data reveal noticeable artifacts. For instance, SPECAT, ESSA, and DCTrans show distinct visual differences between regions with and without spectral observations. Although UTAL mitigates this inconsistency, it yields overly smooth results lacking in fine texture detail. In contrast, both PSTUN and the proposed IrregFusion framework produce visually consistent results across the entire image. Notably, IrregFusion delivers sharper and more structurally faithful reconstructions, underscoring its enhanced capability in addressing real-world irregular spectral fusion challenges.

TABLE V
ABLATION STUDY ON THE EFFECTIVENESS OF THE SPECTRAL PROPAGATION STRATEGY ON THE GF5 DATASET UNDER PARTIAL AND POINT FOV OVERLAP SETTINGS

Partial FoV Overlap				
Methods	Metrics	70%	50%	30%
w/o SPS	PSNR↑	30.71	30.44	30.04
	SAM↓	3.05	3.18	3.36
Ours_SsDRM	PSNR↑	31.28	30.55	30.07
	SAM↓	2.75	3.04	3.25
Point FoV Overlap				
Methods	Metrics	1%	0.5%	0.1%
w/o SPS	PSNR↑	30.27	30.10	29.48
	SAM↓	3.34	3.39	3.64
Ours_SsDRM	PSNR↑	31.36	30.95	29.99
	SAM↓	2.70	2.83	3.23

In conclusion, most comparison methods, when applied using their original training configurations, struggle to effectively handle spatial inconsistencies in practical scenarios, leading to prominent artifacts and performance degradation. In comparison, the proposed IrregFusion framework demonstrates superior reconstruction accuracy and visual fidelity across both synthetic and real datasets, confirming its robustness and effectiveness in both spatially consistent and inconsistent spectra data fusion tasks.

D. Ablation Study

In this section, we conduct three sets of ablation experiments to validate the effectiveness of the key components proposed in the IrregFusion framework: 1) the spectral propagation strategy; 2) the spatial–spectral dual reconstruction architecture; and 3) the self-supervised adaptation mechanism. The following subsections provide a detailed description and analysis of each experiment.

1) Evaluation of Spectral Propagation Strategy: In this article, we conduct a set of ablation experiments to evaluate the effectiveness of the proposed spectral propagation strategy. In particular, we remove this component from the IrregFusion framework, and refer to the resulting variant as "w/o SPS." The quantitative results are reported in Table V, based on the GF5 dataset under both partial spectra and point spectra settings

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TABLE VI

ABLATION STUDY ON THE EFFECTIVENESS OF THE SPATIAL–SPECTRAL DUAL RECONSTRUCTION ARCHITECTURE ON THE GF5 DATASET UNDER PARTIAL AND POINT FOV OVERLAP SETTINGS

Partial FoV Overlap				
Methods	Metrics	70%	50%	30%
w/o Dual	PSNR↑	30.16	29.57	29.00
	SAM↓	3.38	3.63	3.88
Ours_SsDRM	PSNR↑	31.28	30.55	30.07
	SAM↓	2.75	3.04	3.25
Point FoV Overlap				
Methods	Metrics	1%	0.5%	0.1%
w/o Dual	PSNR↑	30.27	30.10	29.48
	SAM↓	3.34	3.39	3.64
Ours_SsDRM	PSNR↑	31.36	30.95	29.99
	SAM↓	2.70	2.83	3.23

TABLE VII

ABLATION STUDY ON THE EFFECTIVENESS OF THE SELF-SUPERVISED ADAPTATION MECHANISM ON THE GF5 DATASET UNDER PARTIAL AND POINT FOV OVERLAP SETTINGS

Partial FoV Overlap				
Methods	Metrics	70%	50%	30%
w/o SsAM	PSNR↑	31.28	30.55	30.07
	SAM↓	2.75	3.04	3.25
Ours	PSNR↑	32.08	31.29	30.85
	SAM↓	2.48	2.75	2.95
Point FoV Overlap				
Methods	Metrics	1%	0.5%	0.1%
w/o SsAM	PSNR↑	31.36	30.95	29.99
	SAM↓	2.70	2.83	3.23
Ours	PSNR↑	32.25	31.71	30.13
	SAM↓	2.38	2.56	3.13

TABLE VIII

SENSITIVITY ANALYSIS OF THE IMPACT OF THE LOSS WEIGHTING PARAMETERS γ, EVALUATED ON THE GF5 DATASET

Partial FoV Overlap (70%)						
Metrics	0	0.01	0.1	0.5	1	10
PSNR↑	31.86	31.93	32.08	31.83	31.77	31.70
SAM↓	2.54	2.52	2.48	2.52	2.53	2.53
Point FoV Overlap (1%)						
PSNR↑	32.16	32.15	32.25	32.17	32.12	32.07
SAM↓	2.42	2.43	2.38	2.42	2.45	2.47

TABLE IX

COMPUTATION COST DETAILS OF EACH METHOD

Methods	ESSA	LGCT	SPECAT	DCTrans	PSTUN
FLOPs (T)	4.136	0.163	0.060	5.016	2.269
Params (M)	6.736	5.778	0.337	8.507	30.22
Methods	UTAL	MCIFNet	CLoRF	OTIAS	Ours
FLOPs (T)	132.500	0.761	3470.000	0.579	106.661
Params (M)	7.373+1.013	2.601	1.326	3.373	52.516+0.789

with varying ratios. Since the spectral propagation strategy primarily influences the performance of the reconstruction module, we focus solely on the results of the reconstruction module (termed as "Ours_SsDRM") in Table V. As observed, removing the spectral propagation strategy leads to a noticeable performance drop, particularly in terms of the SAM metric. This performance degradation suggests that the proposed spectral propagation strategy plays a critical role by aggregating spatially distributed spectral information, thereby alleviating the difficulty of modeling spatial–spectral correlations in subsequent processing stages.

2) Evaluation of Spatial–Spectral Dual Reconstruction Architecture: In this article, we conduct a set of ablation experiments to evaluate the effectiveness of the proposed dual reconstruction architecture. In particular, we remove this component from the IrregFusion framework and denote the resulting variant as "w/o Dual." The quantitative results are summarized in Table VI, based on the GF5 dataset under both partial spectra and point FoV overlap settings with varying ratios. As the dual reconstruction architecture plays a central role in enhancing the performance of the reconstruction module, we report only the results of the reconstruction module (termed as "Ours_SsDRM") in Table VI. From the results, we observe a noticeable performance drop when this component is removed; for instance, the PSNR decreases by approximately 0.5–1 dB. This degradation indicates that the dual reconstruction architecture is essential for effectively capturing spatial–spectral correlations, thereby improving the reconstruction quality under spatial–spectral inconsistency conditions.

3) Evaluation of Self-Supervised Adaptation Mechanism: To assess the effectiveness of the self-supervised adaptation mechanism, we perform an ablation study by removing this component from the IrregFusion framework, referring to the resulting variant as "w/o SsAM." The quantitative results are presented in Table VII, based on the GF5 dataset under both partial spectra and point FoV overlap settings with varying ratios. As shown in the table, excluding the self-supervised adaptation mechanism leads to a clear performance decline; for example, the SAM metric deteriorates by approximately 0.3. This performance drop underscores the importance of the adaptation mechanism in enabling IrregFusion to effectively adapt to challenging scenarios, thereby enhancing reconstruction accuracy in the presence of spatial–spectral inconsistencies.

Moreover, we perform a sensitivity analysis to assess the impact of the loss weighting parameters in (9). Since the first two fidelity terms are equally important, we follow the setting in [12] and set α and β in (9) to 1 by default. Thus, the analysis primarily focuses on the effectiveness of the third term, namely the weak supervisory constraint. To this end, we conduct a series of experiments with varying values of γ, and the corresponding results are reported in Table VIII. When γ is set to 0, the weak supervisory constraint is excluded from (9), leading to noticeable performance degradation. Furthermore, we observe that the best performance is achieved when $\gamma = 0.1$, which is therefore adopted as the default setting in this study.

E. Computation Cost Analysis

Moreover, we further performed a computational cost analysis to evaluate the efficiency of each method. Since the runtime of a single sample on a specific device can be influenced by various external factors, we report the number of floating-

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point operations (FLOPs) and parameters instead. For a fair comparison, Params and FLOPs are calculated with a default image size of 512×512, the detailed results are presented in Table IX. As shown, methods based on self-supervised learning (e.g., UTAL, CLoRF, and the proposed approach) generally incur higher computational costs, as they require multiple iterations to optimize the final fusion results. In particular, CLoRF requires 30 000 iterations; thus, despite having relatively few parameters (1.326 M), it demands up to 3470.000 T FLOPs. In contrast, other methods only need a single forward pass on the pretrained modules, resulting in significantly lower FLOPs. Although the proposed method involves relatively higher FLOPs and parameter counts, it achieves substantial performance improvements over the other approaches.

V. Conclusion

In this study, we presented IrregFusion, a general framework for HSI fusion that effectively addresses the challenge of spatial FoV inconsistency in remote sensing scenarios. Unlike conventional fusion methods that rely on the assumption of complete spatial alignment and dense spectral coverage, IrregFusion is capable of handling both spatially consistent and irregularly distributed spectral data, including partially FoV overlapping HSI and sparsely sampled point spectra. The proposed framework integrates a dual-branch Transformer-based reconstruction module to jointly model intramodal and intermodal correlations between the irregular HSI and the HR MSI. To further mitigate the difficulty of reconstructing spectral information in unobserved regions, we introduce a spectral propagation strategy that diffuses observed spectral information into neighboring regions with missing data. In addition, a self-supervised adaptation mechanism is incorporated to enhance spectral–spatial representation learning and improve generalization performance across various challenging conditions. Extensive experiments conducted on multiple benchmark datasets validate the effectiveness and robustness of IrregFusion.

In addition, to ensure comparability with existing methods and to isolate the influence of FoV inconsistency, this study assumes that the observed LR HSI and HR MSI are spatially aligned within their overlapping FoV regions and are unaffected by degradation. Nonetheless, in real-world scenarios, even the overlapping regions may exhibit misregistration or degradation due to sensor geometry discrepancies, platform motion, or atmospheric interference. Relaxing these assumptions and developing fusion models robust to such imperfections represent valuable directions for future research.

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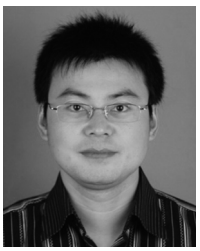
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